

Mathematical Foundations of Political Methodology

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- Proposition
- Methods of Mathematical Proof
- Social Scientific Thinking

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Proposition

Definition

A proposition is a statement that is either true or false.

Now, let's see two examples. The first proposition is true while the second one is wrong.

Proposition: $2 + 3 = 5$.

Proposition: $1 + 1 = 3$.

Remark: Proposition is synonymous with equation!

Proposition: Superstition is the belief in the causal nexus. - Wittgenstein

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Proposition: Not(P)

The truth table for "NOT(P)" is:

P	Not(P)
T	F
F	T

Example. Following proposition is true: Justin is a male. However, not (P), that is, Justin is not a male, is not true.

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Proposition: P and Q

The truth table for "P and Q" is:

P	Q	P and Q
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Example. Say, P, Justin is a male. Q, Justin is a female. P and Q is not true. Because Justin cannot simultaneously be male or female.

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Proposition: P or Q

The truth table for "P or Q" is:

P	Q	P or Q
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F	T	T
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Here, let's ignore the discussion of how we should categorize gender, and assume that there are only two genders.

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The truth table for " P implies Q " is:

P	Q	P implies Q
T	T	T
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The truth table can be summarized in words as: An implication is true exactly when the if-part is false or the then-part is true.

Example 1. If $x > 2$, then, $x + 6 > 8$. P is true, Q is true, thus, P implies Q is true.

Example 2. If pigs can fly, then, dogs can drive a car. Since P is false, thus, P implies Q is true.

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Proposition: $P \text{ IFF } Q$

The truth table for $P \text{ IFF } Q$ (if and only if) Q is:

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Example. This if-and-only-if statement is true for every real number x :
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Proposition: Contrapositive

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"NOT(Q) IMPLIES NOT(P)" is called the contrapositive of the implication "P IMPLIES Q". These two are just different ways of saying the same thing.

Their truth table is:

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Example. "If I am hungry, then I am grumpy." is equivalent to "If I am not grumpy, then I am not hungry".

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Proposition: Always vs. Sometimes True

Examples on "Always True":

For all $x \in \mathbb{R}$, $x^2 \geq 0$. Equivalent to write as: $\forall x \in \mathbb{R}, x^2 \geq 0$.

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There exists an $x \in \mathbb{R}$ such that $5x^2 - 7 = 0$. Equivalent to write as:
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Mathematical Proofs

Before we proceed to the definition of mathematical proofs, we need to define axioms first.

Definition

Propositions that are simply accepted as true are called axioms.

Example on axiom: There is a straight line segment between every pair of points.

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Mathematical Proofs

Definition

A mathematical proof of a proposition is a chain of logical deductions leading to the proposition from a base set of axioms.

Mathematical Proofs

Following three concepts are important for proofs.

- Important propositions are called theorems.
- A lemma is a preliminary proposition useful for proving later propositions.
- A corollary is a proposition that follows in just a few logical steps from a lemma or a theorem.

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Methods for Mathematical Proofs

A fundamental logical deduction is modus ponens. It says that a proof of P together with a proof that P IMPLIES Q is a proof of Q .

$$\frac{P, \quad P \text{ implies } Q}{Q}$$

An example of an argument that fits the form modus ponens:

If today is Tuesday, then John will go to work.

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Methods for Mathematical Proofs: Direct Proof

Now, let's see how we do mathematical proofs.

For this method, in order to prove that P IMPLIES Q , there are two steps:
1. Assume P is true; 2. Show that Q logically follows.

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Theorem

If $0 \leq x \leq 2$, then, $-x^3 + 4x + 1 > 0$.

Proof.

Assume $0 \leq x \leq 2$. Then, x , $2 - x$, and $2 + x$ are all nonnegative. Therefore, the product of these terms is also nonnegative. Adding 1 to this product gives a positive number, so:

$$x(2 - x)(2 + x) + 1 \geq 0$$

Which is equivalent to:

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Let A and B be sets. If $A = B$ then $A \subset B$ and $B \subset A$

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Suppose $A = B$. By definition, if $x \in A$ then $x \in B$. So $A \subset B$. Again, by definition, if $y \in B$ then $y \in A$. So $B \subset A$. □

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Methods of Mathematical Proof: Prove the Contrapositive

If prove the contrapositive is easier, we can prove the contrapositive, since it's logically equivalent to the original statement.

To prove the contrapositive, there are two steps: 1. write, "We prove the contrapositive:", and then state the contrapositive; 2. proceed as in the first method.

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If r is irrational, then $\sqrt{r}$ is also irrational.

Proof.

We prove the contrapositive: if $\sqrt{r}$ is rational, then, r is rational. Assume that $\sqrt{r}$ is rational. Then, there exists integers a and b such that:

$$\sqrt{r} = \frac{a}{b}$$

Squaring both sides gives:

$$r = \frac{a^2}{b^2}$$

Since a^2 and b^2 are integers, r is also rational. □

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In a proof by contradiction, you show that if a proposition were false, then some false fact would be true. Since a false fact can't be true, the proposition had better not be false. That is, the proposition really must be true.

The step of proof by contradiction is: 1. Write, "We use proof by contradiction."; 2. Write, "Suppose P is false"; 3. Deduce something known to be false (a logical contradiction); 4. Write, "This is a contradiction. Therefore, P must be true."

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Theorem

There is no largest even integer.

Proof.

Suppose not. That is, suppose that there is a largest even integer. Let's call it k .

Since k is even, it has the form $2n$, where n is an integer. Consider $k + 2$.

$$k + 2 = (2n) + 2 = 2(n + 1).$$

So $k + 2$ is even.

But $k + 2$ is larger than k . This contradicts our assumption that k is the largest even integer.

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Let A and B be sets. Then $A = B$ if and only if $A \subset B$ and $B \subset A$.

Proof.

$\Rightarrow$ Suppose $A = B$. By definition, if $x \in A$, $x \in B$. So $A \subset B$. Again, by definition, if $y \in B$ then $y \in A$. So $B \subset A$.

$\Leftarrow$ Suppose $A \subset B$ and that $B \subset A$. Now, by way of contradiction, suppose that $A \neq B$. $A \neq B$ only if there is $x \in A$ and $x \notin B$ or if $y \in B$ and $y \notin A$. But then, either $A \not\subset B$ or $B \not\subset A$, contradicting our initial assumption. □

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- Proposition
- Methods of Mathematical Proof
- Social Scientific Thinking

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Example. The statement, "capitalism exploits the poor" is not valid unless you give enough evidence to support the statement.

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